



Stroke

Stroke is an emergency condition that occurs when the blood flow to the brain is suddenly interrupted. The blood vessel either becomes blocked or ruptures. Brain cells cannot receive oxygen and nutrients and die within a short period of time. This leads to impairment of functions such as speech, movement, vision, or balance.

There are three main types of stroke: 1) Ischemic stroke: The most common type. A blood vessel in the brain becomes blocked by a clot or narrowing. 2) Hemorrhagic stroke: A blood vessel in the brain ruptures, causing bleeding. 3) Transient ischemic attack (TIA): Blood flow is interrupted for a short period of time. The symptoms usually resolve within a few minutes or hours and do not cause permanent damage. However, it may be a warning sign of a future major stroke.

In 2021, there were approximately 93.8 million people living with stroke worldwide, while 11.9 million new cases were reported in the same year. Stroke is one of the leading causes of death and disability. Approximately one in four adults is at risk of experiencing a stroke during their lifetime. This risk has increased by 50% over the past 20 years.

The burden of stroke (years lost due to death and disability) has also increased. A large proportion of deaths and disabilities occur particularly in low- and middle-income countries. The global economic cost of stroke is estimated to exceed 890 billion dollars.

Symptoms of stroke usually appear suddenly. The most common symptoms include: drooping of one side of the face, weakness in an arm or leg, impaired or unintelligible speech, sudden loss of balance or dizziness, and sudden loss of vision.

Symptoms vary depending on which part of the brain is affected. The right hemisphere controls the left side of the body, while the left hemisphere controls the right side. The location and extent of the damage determine how severely the person will be affected. Some strokes do not cause obvious symptoms; these are referred to as “silent strokes.” They may increase the risk of dementia.

The symptoms of TIA are short-lived and completely resolve. Nevertheless, it still requires urgent medical attention.

The most important modifiable risk factor for stroke is high blood pressure. In people with high blood pressure, the risk of stroke is nearly three times higher. Other important factors include: smoking, high cholesterol, diabetes, obesity and physical inactivity, a diet high in salt, harmful alcohol use, air pollution, and kidney problems.

Non-modifiable factors include age- after the age of 55, the risk doubles every ten years- previous stroke, and certain heart conditions, such as atrial fibrillation. The prevalence of atrial fibrillation increases with age, and it accounts for approximately 20–25% of strokes in patients over 80 years of age.

Stroke is less common in young adults (18-55 years) but its causes are different. In this age group, arterial dissection, a cardiac opening (patent foramen ovale), certain blood-clotting disorders, or rare forms of vascular inflammation are more frequently involved. In older adults, high blood pressure, atherosclerosis, and heart-derived blood clots are the main contributing factors.

Brain cells require a continuous blood supply. When blood flow is interrupted, the cells die. In ischemic stroke, a blood vessel is blocked by a clot or plaque. The clot may form in the brain or may come from another part of the body. In hemorrhagic stroke, a weakened blood vessel ruptures; high blood pressure gradually weakens blood vessels over time.

Different areas of the brain control different functions. Depending on the area that is damaged, speech, movement, vision, or thinking may be affected.

If stroke is suspected, the patient should be taken to the hospital immediately. Brain imaging, such as CT or MRI, is

performed for diagnosis. This determines whether the cause is bleeding or blockage. In young patients, additional investigations may be required to identify the underlying cause.

Time is particularly critical. In ischemic stroke, clot-dissolving medication (for example, tPA) may be administered during the first hours after symptom onset. In some patients, a procedure to remove the clot from the blood vessel (thrombectomy) may be performed. In hemorrhagic stroke, blood pressure is controlled, and surgery may be considered when necessary.

Patients have better outcomes when they are monitored in specialized stroke units. Once the patient is stable, rehabilitation begins, including physical therapy, speech therapy, activities of daily living training, and psychological support.

In the first days after a stroke, complications such as brain edema, difficulty swallowing, pneumonia, seizures, or shoulder pain may occur. In the long term, permanent limitations in movement, speech disorders, memory problems, depression, and loss of independence may develop. Stroke increases the risk of dementia by more than twofold. Silent strokes may also contribute to dementia.

In young patients, standard approaches may not be sufficient to identify the cause of stroke; specialized investigations may be required. In addition, the relationship between stroke and dementia is increasingly better understood. Controlling common risk factors- such as high blood pressure, high cholesterol, diabetes, and physical inactivity- may help reduce the risk of both stroke and dementia.

In conclusion, stroke is a largely preventable condition. Controlling high blood pressure, quitting smoking, maintaining a healthy diet, and engaging in physical activity are the most effective steps. Recognizing the symptoms and seeking immediate medical assistance increase the chances of survival and recovery.

References:

- 1) World Health Organization. (2025, December 19). Stroke. <https://www.who.int/news-room/fact-sheets/detail/stroke>
- 2) Murphy SJ, Werring DJ. Stroke: causes and clinical features. *Medicine* (Abingdon). 2020 Sep;48(9):561-566. doi: 10.1016/j.mpmed.2020.06.002. Epub 2020 Aug 6. PMID: 32837228; PMCID: PMC7409792.
- 3) Feigin VL, Brainin M, Norrving B, Martins SO, Pandian J, Lindsay P, F Grupper M, Rautalin I. World Stroke Organization: Global Stroke Fact Sheet 2025. *Int J Stroke*. 2025 Feb;20(2):132-144. doi: 10.1177/17474930241308142. Epub 2025 Jan 3. PMID: 39635884; PMCID: PMC11786524.

4) Heart and Stroke Foundation. (n.d.). Stroke and the brain. <https://www.heartandstroke.ca/sea/rch-results-page?q=stroke>

5) Krüger, C. (2026, April 10). Childhood ischemic stroke: Why young patients need a different diagnostic approach. ESO / ESOC / Stroke Research <https://eso-stroke.org/juvenile-ischaemic-stroke-why-young-patients-require-a-different-diagnostic-mindset/>

Publication Date: September 1, 2026

Corresponding Author: Ali Bocakkıtaç/ Medical Student, Faculty of Medicine, Selçuk University